Calculus I Lesson 39C

1) $f(x)=\ln(3x)$.

a) $\ln(3x)=\ln(\underline{\quad ? \quad})+\ln(\underline{\quad ? \quad})$

b) $f'(x)=\ D_x\left[\ln(3x)\right]=\underline{\quad ? \quad}$

2) $f(x)=\ln\left(x^4\right)$.

a) $\ln\left(x^4\right)=4\cdot\ln(\underline{\quad ? \quad})$

b) $f'(x)=\ D_x\left[\ln\left(x^4\right)\right]=\underline{\quad ? \quad}$

3) $f(x) = \ln\left(x^2 + 4\right)$.

Hint: Use the Chain Rule.

a) Let $u = x^2 + 4$. $\dfrac{du}{dx} = \underline{\quad ? \quad}$

b) $D_x\left[\ln\left(x^2 + 4\right)\right] = D_x\left[\ln(u)\right] = \dfrac{1}{u} \cdot \dfrac{du}{dx} = \underline{\quad ? \quad}$

4) $f(x) = \ln\left(2x^2 + 4x + 3\right)$.

Hint: Use the Chain Rule.

a) Let $u = 2x^2 + 4x + 3$. $\dfrac{du}{dx} = \underline{\quad ? \quad}$

b) $D_x\left[\ln\left(2x^2 + 4x + 3\right)\right] = D_x\left[\ln(u)\right] = \dfrac{1}{u} \cdot \dfrac{du}{dx} = \underline{\quad ? \quad}$

5) $f(x) = \ln\left(x^3 + 3x^2 + 1\right)$.

Hint: Use the Chain Rule.

a) Let $u = x^3 + 3x^2 + 1$. $\dfrac{du}{dx} = \underline{\quad ? \quad}$

b) $D_x\left[\ln\left(x^3 + 3x^2 + 1\right)\right] = D_x\left[\ln(u)\right] = \dfrac{1}{u} \cdot \dfrac{du}{dx} = \underline{\quad ? \quad}$

6) $f(x) = \ln\left(\sqrt{x^2+5}\right)$.

Hint: $\ln\left(\sqrt{x^2+5}\right) = \ln\left(x^2+5\right)^{1/2} = \frac{1}{2}\cdot\ln\left(x^2+5\right)$

a) Let $u = x^2+5$. $\frac{du}{dx} = \underline{\quad ? \quad}$

b) $D_x\left[\ln\left(\sqrt{x^2+5}\right)\right] = D_x\left[\frac{1}{2}\ln\left(x^2+5\right)\right] = \frac{1}{2}\cdot D_x\left[\ln\left(x^2+5\right)\right] = \frac{1}{2}\cdot D_x\left[\ln(u)\right] = \frac{1}{2}\cdot\frac{1}{u}\cdot\frac{du}{dx} = \underline{\quad ? \quad}$

7) $f(x) = \ln\left(\sqrt{x^3 + 2x + 1}\right)$.

Hint: $\ln\left(\sqrt{x^3 + 2x + 1}\right) = \ln\left(x^3 + 2x + 1\right)^{1/2} = \frac{1}{2} \cdot \ln\left(x^3 + 2x + 1\right)$

a) Let $u = x^3 + 2x + 1$. $\frac{du}{dx} = \underline{\quad ? \quad}$

b) $D_x\left[\ln\left(\sqrt{x^3 + 2x + 1}\right)\right] = D_x\left[\frac{1}{2}\ln\left(x^3 + 2x + 1\right)\right] = \frac{1}{2} \cdot D_x\left[\ln\left(x^3 + 2x + 1\right)\right]$

$$= \frac{1}{2} \cdot D_x\left[\ln(u)\right] = \frac{1}{2} \cdot \frac{1}{u} \cdot \frac{du}{dx} = \underline{\quad ? \quad}$$

8) $f(x) = \ln\left(x \cdot \sqrt{x^3 + 2x + 1}\right)$.

Hint: $\ln\left(x \cdot \sqrt{x^3 + 2x + 1}\right) = \ln x + \ln\left(\sqrt{x^3 + 2x + 1}\right)$

$D_x\left[\ln\left(x \cdot \sqrt{x^3 + 2x + 1}\right)\right] = \underline{\quad ? \quad}$

9) $f(x) = \ln\left(\sqrt{\frac{x+12}{x-12}}\right)$

a) $\ln\left(\sqrt{\frac{x+12}{x-12}}\right) = \ln\left(\frac{x+12}{x-12}\right)^{1/2} = \frac{1}{2}\cdot\left[\ln\left(\frac{x+12}{x-12}\right)\right] = \frac{1}{2}\cdot\left[\ln(\underline{\quad ? \quad}) - \ln(\underline{\quad ? \quad})\right]$

b) $D_x\left[\ln\left(\sqrt{\frac{x+12}{x-12}}\right)\right] = \underline{\quad ? \quad}$

10) $f(x)=\ln\left(\sqrt{1+\cos x}\right)$.

Find the tangent line at the point $\left(\dfrac{\pi}{2},0\right)$

Hint: $\ln\left(\sqrt{1+\cos x}\right)=\ln(1+\cos x)^{1/2}=\dfrac{1}{2}\cdot\ln(1+\cos x)$

a) Let $u=1+\cos x$. $\dfrac{du}{dx}=$ ___?___

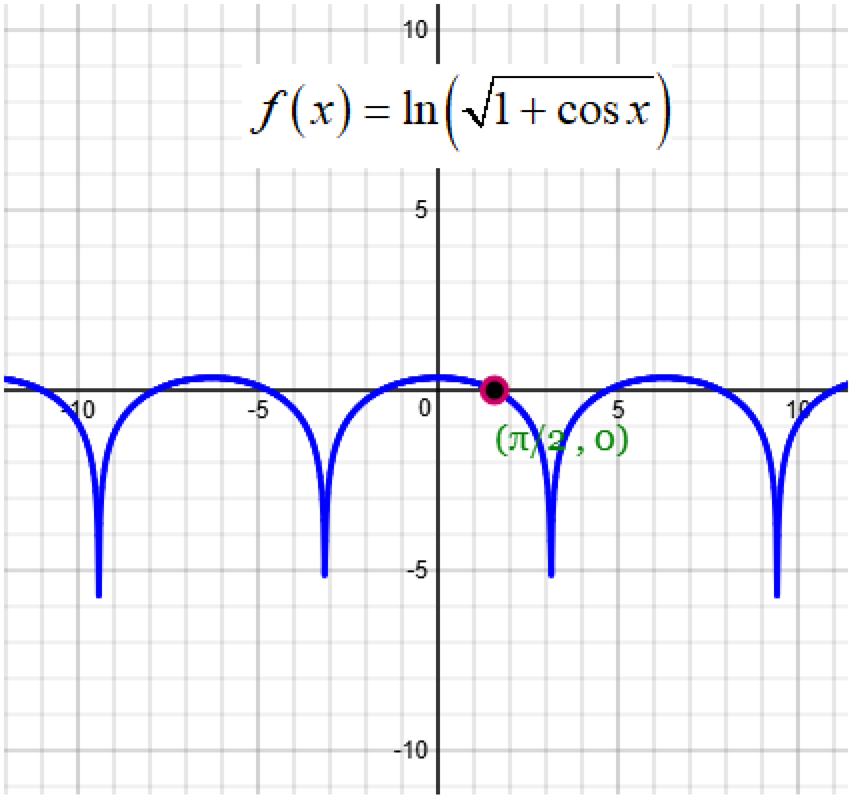
b) $f'(x)=D_x\left[\ln\left(\sqrt{1+\cos x}\right)\right] = D_x\left[\dfrac{1}{2}\cdot\ln(1+\cos x)\right]$

$$=\frac{1}{2}\cdot D_x\left[\ln(1+\cos x)\right]=\frac{1}{2}\cdot D_x\left[\ln(u)\right]=\frac{1}{2}\cdot\frac{1}{u}\cdot\frac{du}{dx}=\underline{\quad ?\quad}$$

c) Slope of tangent line = $f'\left(\dfrac{\pi}{2}\right)=$ ___?___

d) Equation of tangent line is ___?___

Hint: $y-y_1=m(x-x_1)$



$f(x) = \ln\left(\sqrt{1+\cos x}\right)$
10
5
-10
-5
0
5
10
-5
-10
(π/2 , 0)

11) $f(x)=\ln\left((x+1)^3\right)$.

Find the tangent line at the point $(2,\ 3\ln 3)$

Hint: $\ln\left((x+1)^3\right)=3\cdot\ln(x+1)$

a) Let $u=x+1$. $\dfrac{du}{dx}=$ ___?___

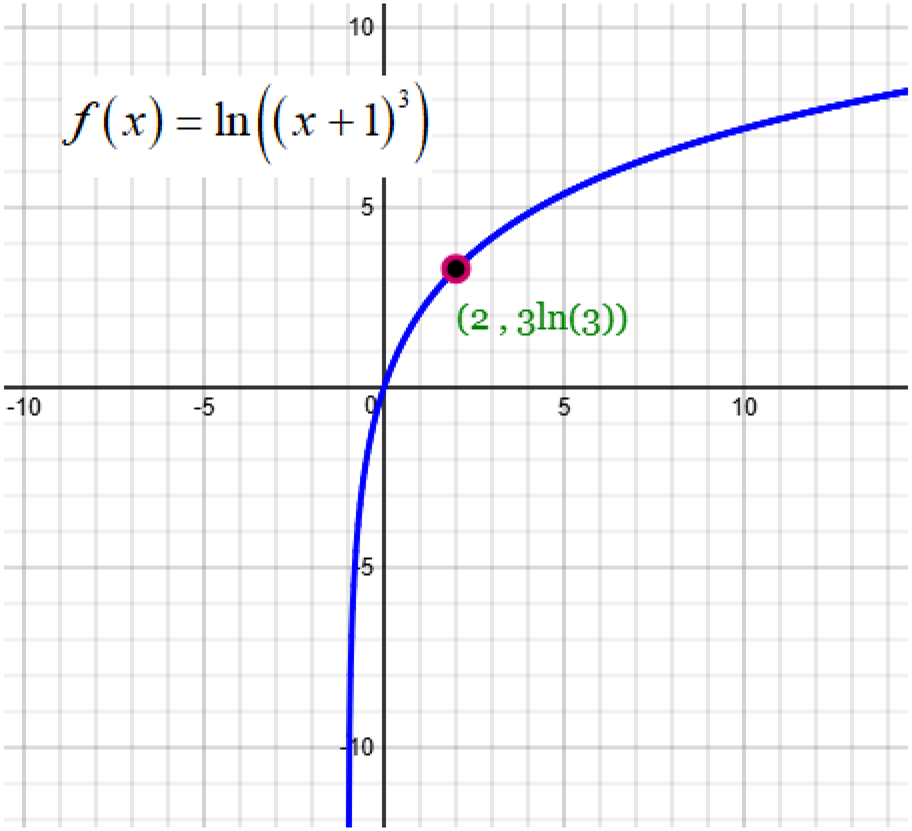
b) $f'(x)=D_x\left[\ln\left((x+1)^3\right)\right]=D_x\left[3\cdot\ln(x+1)\right]$

$$=3\cdot D_x\left[\ln(x+1)\right]=3\cdot D_x\left[\ln(u)\right]=3\cdot\frac{1}{u}\cdot\frac{du}{dx}=\text{\_\_\_?\_\_\_}$$

c) Slope of tangent line $=f'(2)=$ ___?___

d) Equation of tangent line is ___?___

Hint: $y-y_1=m(x-x_1)$



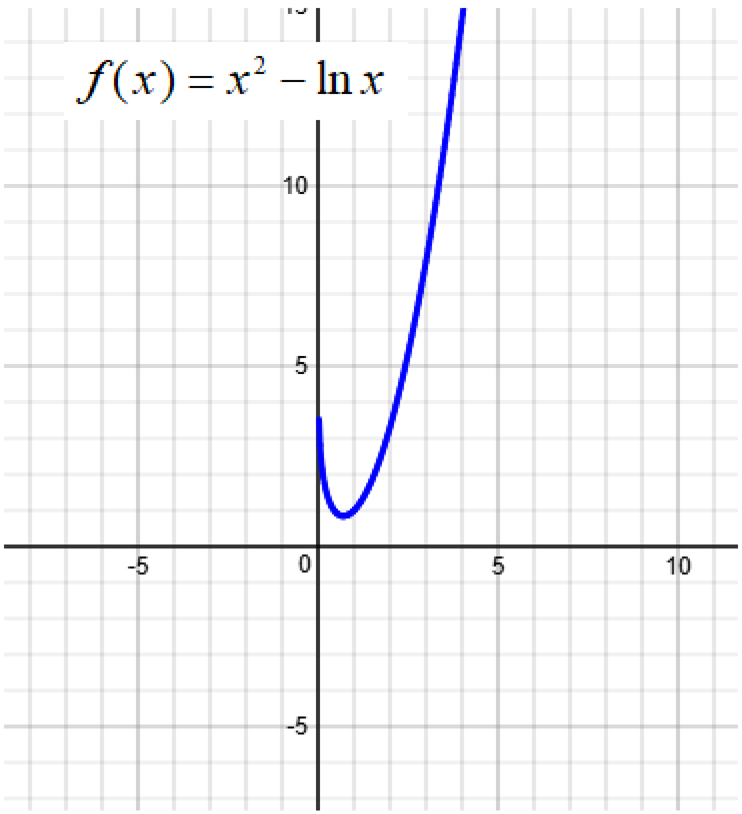
$f(x) = \ln\left((x+1)^3\right)$
(2 , 3ln(3))
10
5
-5
-10
-10
-5
0
5
10

12) $f(x) = x^2 - \ln x$

a) $f'(x) =$ ___?___

b) Set $f'(x) = 0$ and solve for x. $x =$ ___?___

c) The function $f(x) = x^2 - \ln x$ has a minimum at the point ___?___.



13) $f(x) = x \cdot \ln x$

a) $f'(x) = x \cdot D_x\left[\ln(x)\right] + \ln x \cdot D_x\left[x\right] =$ ___?___

b) Set $f'(x) = 0$ and solve for x. $x =$ ___?___

c) The function $f(x) = x \cdot \ln x$ has a minimum at the point ___?___.